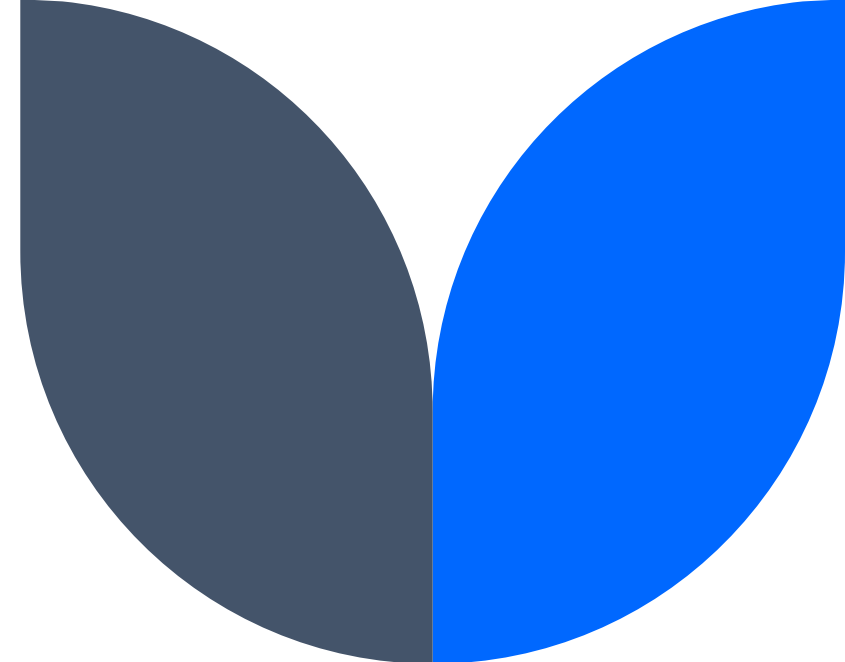
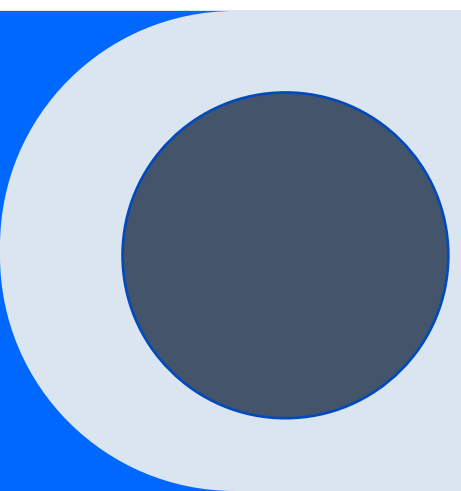




Low Power Configurable Multi Clock Digital System From RTL to GDS



By: Youssef Khaled

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RTL Design

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Logic Synthesis using Design Compiler

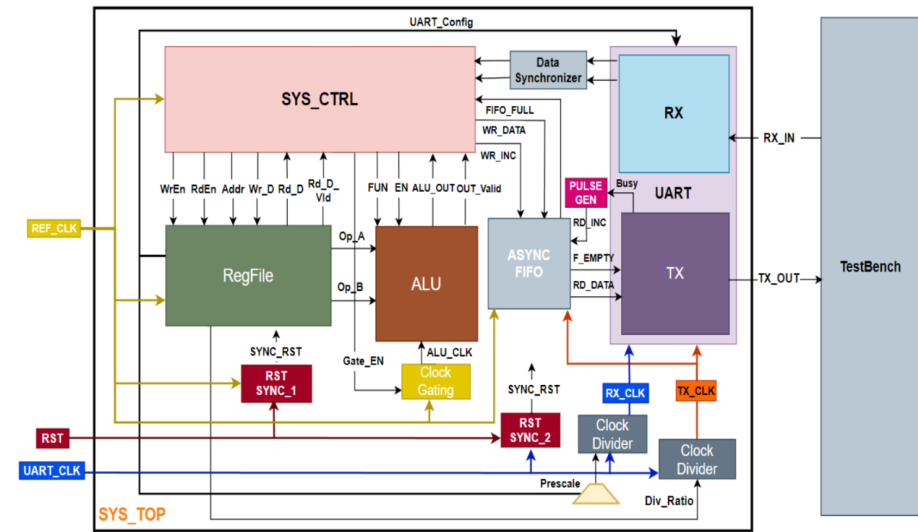
Post-Synthesis Formal Verification

DFT Insertion

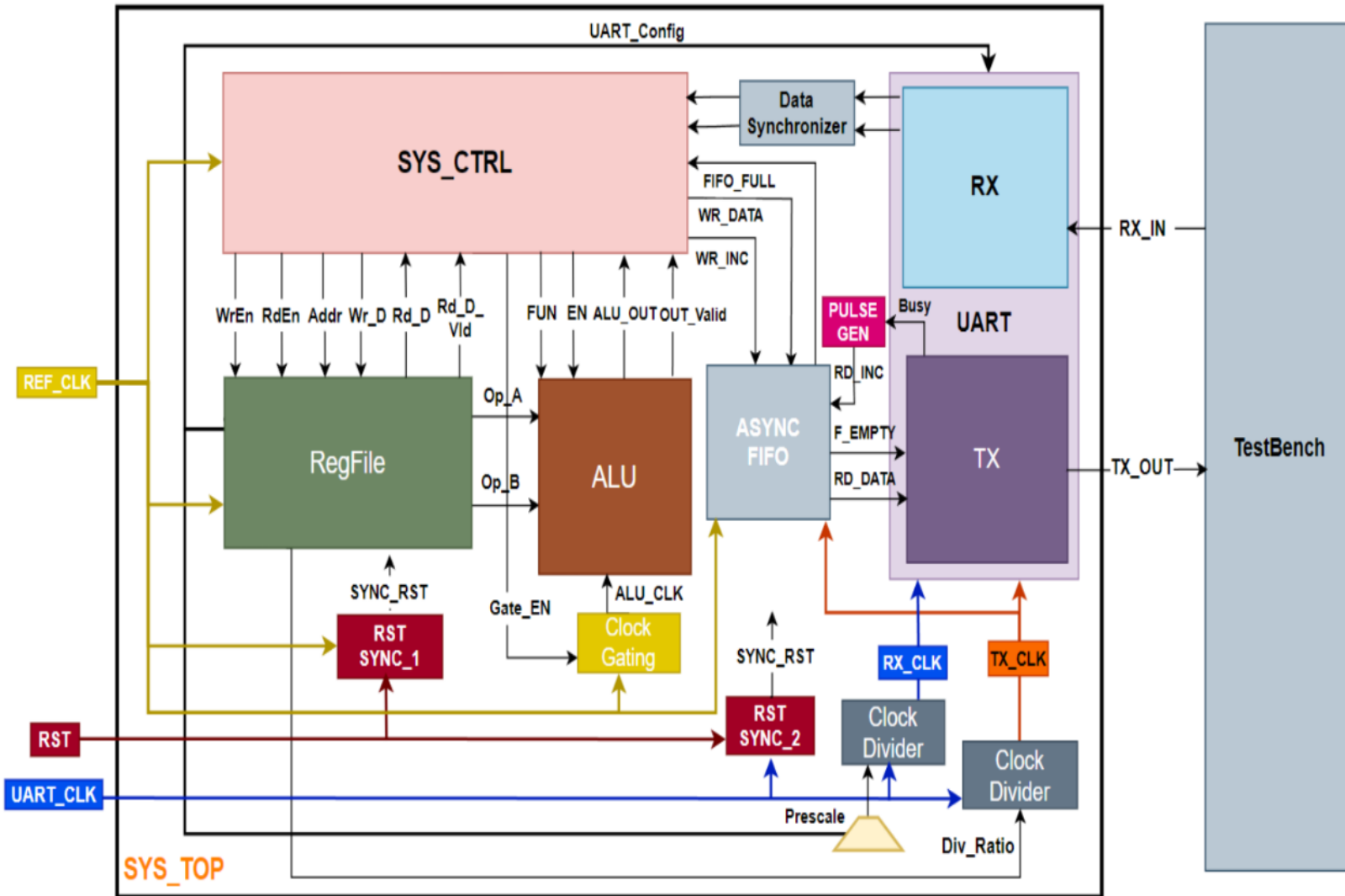
Post-DFT Formal Verification

PnR Flow

Introduction



Block Diagram



System Overview

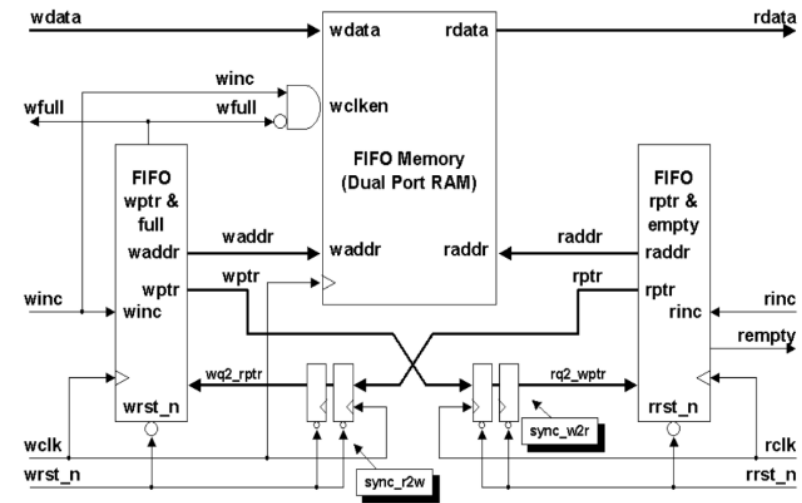
- The system is responsible of receiving commands through UART receiver to do different system functions as register file reading/writing or doing an ALU operation.
- Result is then sent to UART transmitter through asynchronous FIFO for handling different clock rates and avoid data loss.
- The system uses clock gating to minimize dynamic power consumed by the ALU.
- The system uses clock dividers to get the proper pre-scale for UART RX and proper clock for UART TX.
- Supported ALU operations are Addition, Subtraction, Multiplication, Division, Shift left by one bit, Shift right by one bit, logical operations, equality, greater-than comparison.
- Supported register file operators are read and write.



System Specifications

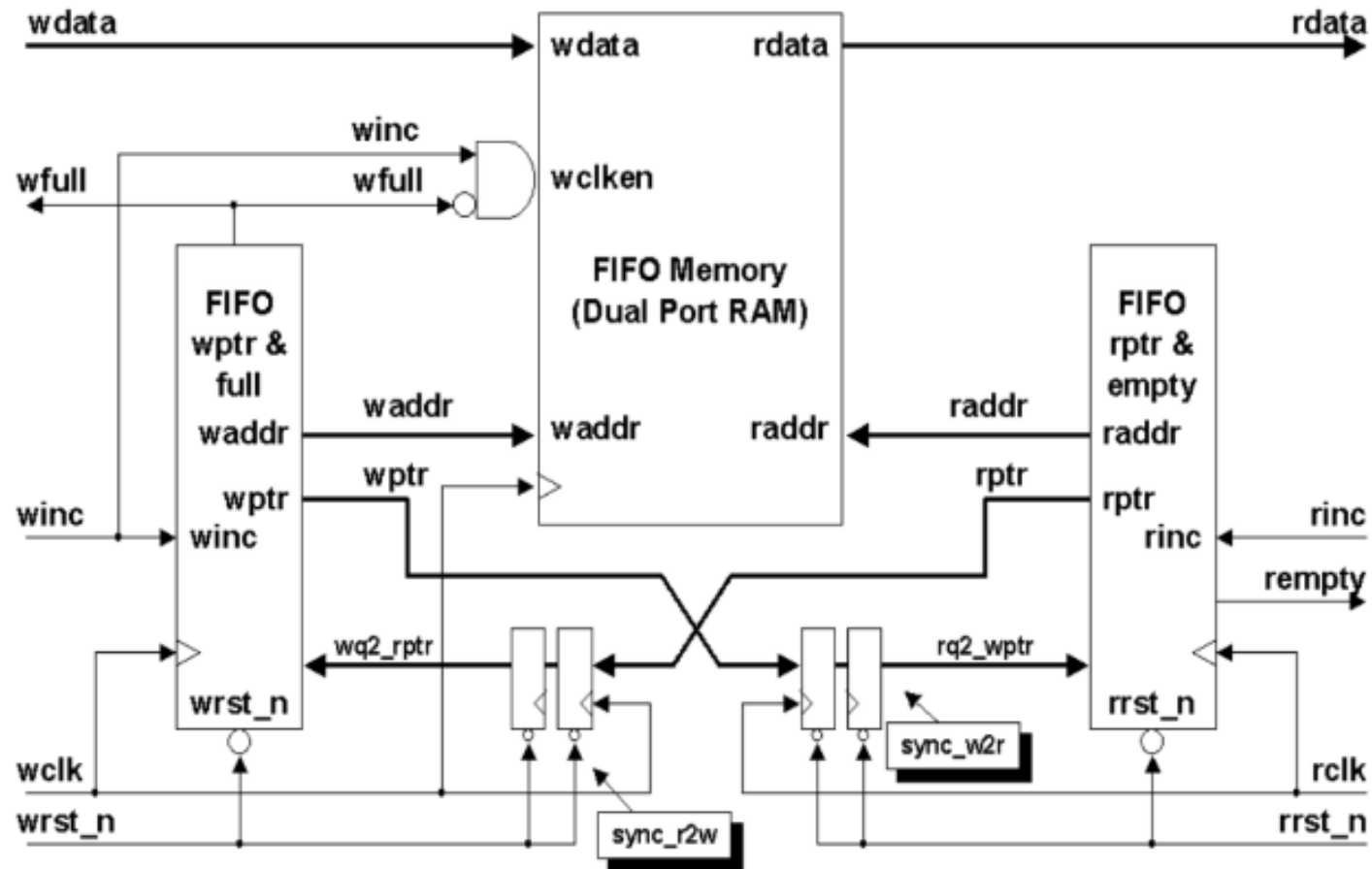
- Reference clock frequency is 50 MHz.
- UART clock frequency is 3.68 MHz.
- Clock dividers are always on.
- System supports 4 commands:
 - 1 – Register file write. (3 frames)
 - 2 – Register file read. (2 frames)
 - 3 – ALU operation with input operands. (4 frames)
 - 4 – ALU operation with default operands. (2 frames)

RTL Design

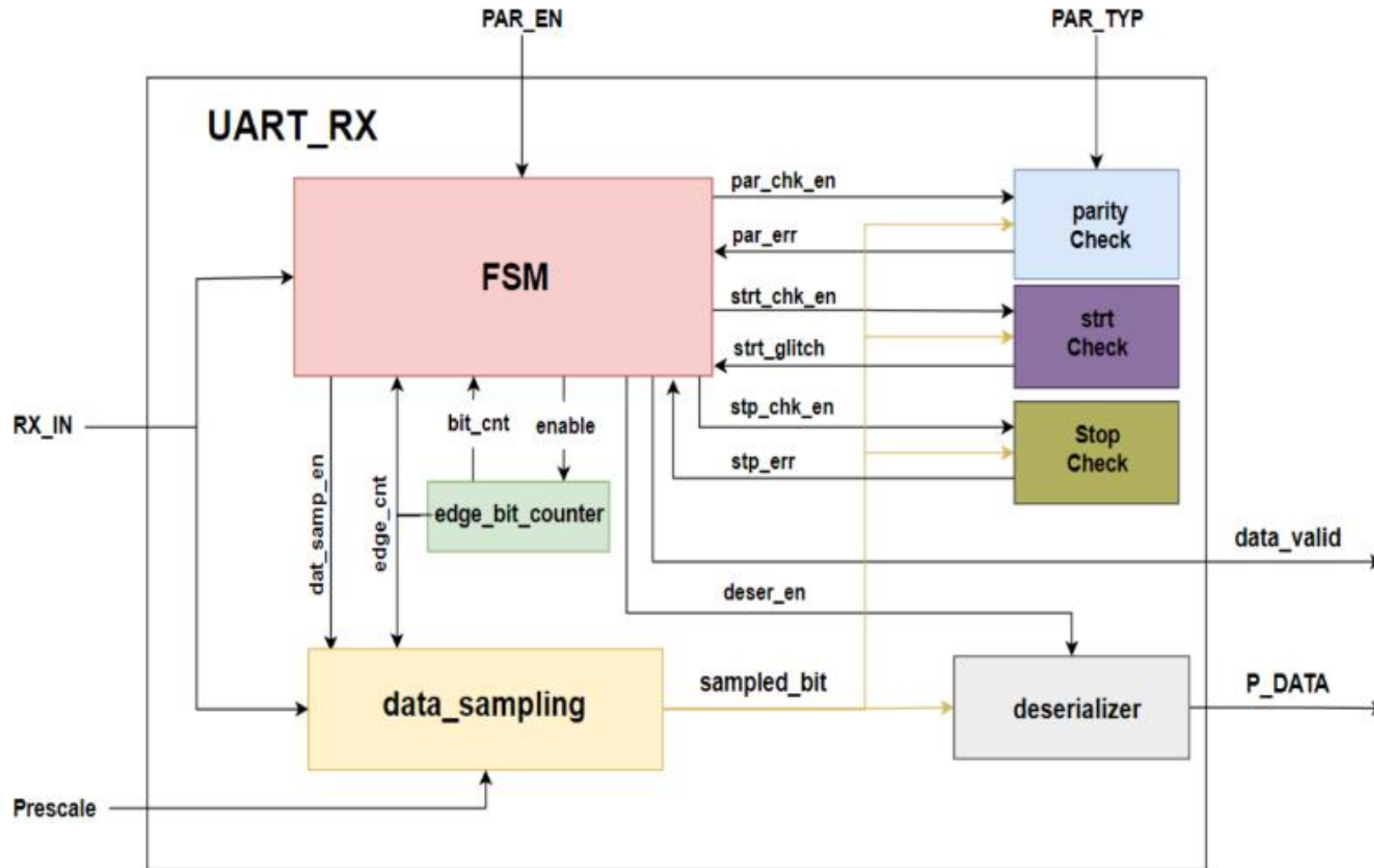


Refer to GitHub [Repo](#) for more details.

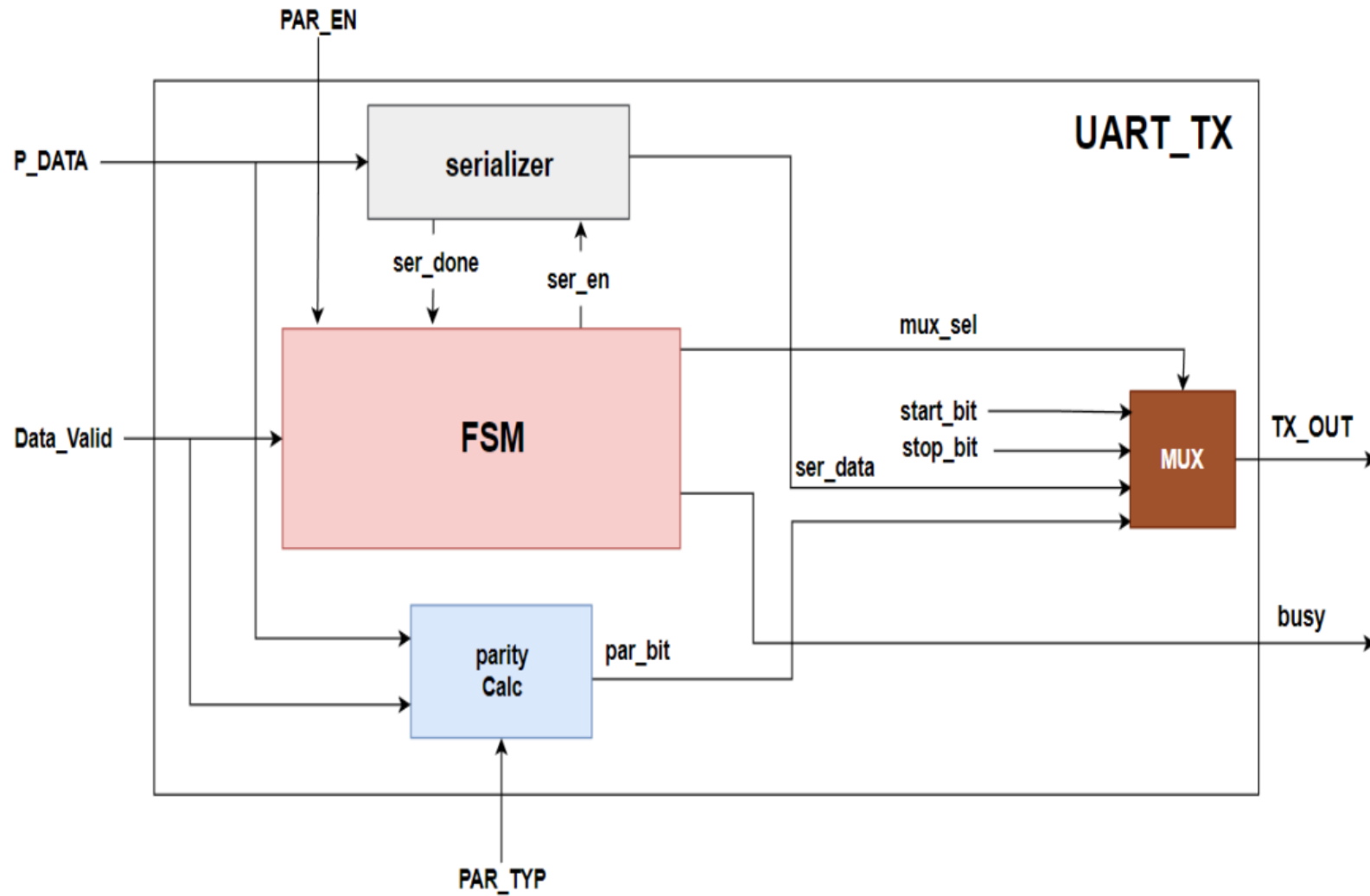
Main Blocks: FIFO



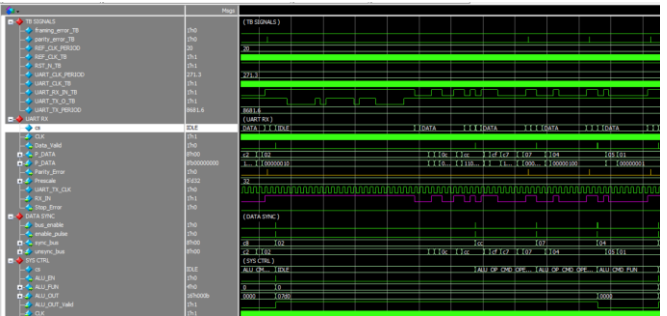
Main Blocks: UART RX



Main Blocks: UART TX



Functional Verification



Testbench Result

```
# ***** Testing with even parity *****  
# Register File Read number 1 succeed.  
# Register File Read number 2 succeed.  
# ALU OPERATION number 1 succeed.  
# ALU OPERATION number 2 succeed.  
# ALU OPERATION number 3 succeed.  
# ALU OPERATION number 4 succeed.  
# ALU OPERATION number 5 succeed.  
# ** Note: $stop      : SYS_TOP_TB.sv(47)  
#      Time: 5317480 ns Iteration: 1 Instance: /SYS_TOP_TB  
# Break in Module SYS_TOP_TB at SYS_TOP_TB.sv line 47
```

VSIM 3>

Logic Synthesis



Constraints Snippet

```
73 create_clock -name $UART_CLK_NAME -period $UART_CLK_PER -waveform "0 [expr $UART_CLK_PER/2]" [get_ports UART_CLK]
74 set_clock_uncertainty -setup $CLK_SETUP_SKEW [get_clocks $UART_CLK_NAME]
75 set_clock_uncertainty -hold $CLK_HOLD_SKEW [get_clocks $UART_CLK_NAME]
76 set_clock_transition -rise $CLK_RISE [get_clocks $UART_CLK_NAME]
77 set_clock_transition -fall $CLK_FALL [get_clocks $UART_CLK_NAME]
78 set_clock_latency $CLK_LAT [get_clocks $UART_CLK_NAME]
79
80
81 create_clock -name $scan_clk_NAME -period $scan_clk_PER -waveform "0 [expr $scan_clk_PER/2]" [get_ports scan_clk]
82 set_clock_uncertainty -setup $CLK_SETUP_SKEW [get_clocks $scan_clk_NAME]
83 set_clock_uncertainty -hold $CLK_HOLD_SKEW [get_clocks $scan_clk_NAME]
84 set_clock_transition -rise $CLK_RISE [get_clocks $scan_clk_NAME]
85 set_clock_transition -fall $CLK_FALL [get_clocks $scan_clk_NAME]
86 set_clock_latency $CLK_LAT [get_clocks $scan_clk_NAME]
87
88
89 create_generated_clock -master_clock $REF_CLK_NAME -source [get_ports REF_CLK] \
90     -name $ALU_CLK_NAME [get_port CLK_GATE_U0/GATED_CLK] \
91     -divide_by 1
92 set_clock_uncertainty -setup $CLK_SETUP_SKEW [get_clocks $ALU_CLK_NAME]
93 set_clock_uncertainty -hold $CLK_HOLD_SKEW [get_clocks $ALU_CLK_NAME]
94 set_clock_transition -rise $CLK_RISE [get_clocks $ALU_CLK_NAME]
95 set_clock_transition -fall $CLK_FALL [get_clocks $ALU_CLK_NAME]
96 set_clock_latency $CLK_LAT [get_clocks $ALU_CLK_NAME]
97
98 create_generated_clock -master_clock $UART_CLK_NAME -source [get_ports UART_CLK] \
99     -name $RX_CLK_NAME [get_port Clk_Div_U1/o_div_clk] \
100     -divide_by 1
101 set_clock_uncertainty -setup $CLK_SETUP_SKEW [get_clocks $RX_CLK_NAME]
102 set_clock_uncertainty -hold $CLK_HOLD_SKEW [get_clocks $RX_CLK_NAME]
103 set_clock_transition -rise $CLK_RISE [get_clocks $RX_CLK_NAME]
104 set_clock_transition -fall $CLK_FALL [get_clocks $RX_CLK_NAME]
105 set_clock_latency $CLK_LAT [get_clocks $RX_CLK_NAME]
106
107 create_generated_clock -master_clock $UART_CLK_NAME -source [get_ports UART_CLK] \
108     -name $TX_CLK_NAME [get_port Clk_Div_U0/o_div_clk] \
109     -divide_by 32
```



Timing Reports: Setup

```
1
2 *****
3 Report : timing
4     -path full
5     -delay max
6     -max_paths 20
7 Design : SYS_TOP
8 Version: K-2015.06
9 Date   : Sat Oct 26 06:57:38 2024
10 *****
11
12 Operating Conditions: scmetro_tsmc_cl013g_rvt_ss_1p08v_125c  Library: scmetro_tsmc_cl013g_rvt_ss_1p08v_125c
13 Wire Load Model Mode: top
14
15 Startpoint: RegFile_U0/regArr_reg[1][6]
16             (rising edge-triggered flip-flop clocked by REF_CLK)
17 Endpoint:  ALU_U0/ALU_OUT_reg[0]
18             (rising edge-triggered flip-flop clocked by ALU_CLK)
19 Path Group: ALU_CLK
20 Path Type: max
21
22 Point                Incr      Path
23 -----
24 clock REF_CLK (rise edge)          0.00      0.00
25 clock network delay (ideal)        0.00      0.00
26 RegFile_U0/regArr_reg[1][6]/CK (DFFRQX2M) 0.00      0.00 r
27 RegFile_U0/regArr_reg[1][6]/Q (DFFRQX2M)  0.48      0.48 f
28 RegFile_U0/REG1[6] (RegFile WIDTH8_DEPTH16_ADDR4) 0.00      0.48 f
29 ALU_U0/B[6] (ALU_OPER_WIDTH8_OUT_WIDTH16) 0.00      0.48 f
30 ALU_U0/U90/Y (BUFEX2M)              0.21      0.70 f
31 ALU_U0/div_52/b[6] (ALU_OPER_WIDTH8_OUT_WIDTH16_DW_div_0) 0.00      0.70 f
32 ALU_U0/div_52/U70/Y (NOR2X1M)        0.16      0.86 r
33 ALU_U0/div_52/U67/Y (AND3X1M)        0.20      1.06 r
34 ALU_U0/div_52/U65/Y (AND2X1M)        0.16      1.22 r
35 ALU_U0/div_52/U62/Y (AND4X1M)        0.25      1.47 r
36 ALU_U0/div_52/U40/Y (CLKMX2X2M)      0.24      1.72 f
37 ALU_U0/div_52/u_div/u_fa_PartRem_0_6_1/CO (ADDFX2M) 0.44      2.15 f
38 ALU_U0/div_52/U63/Y (AND3X1M)        0.32      2.47 f
39 ALU_U0/div_52/U46/Y (CLKMX2X2M)      0.24      2.71 r
40
"violate" not found
```

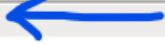
Desi

Timing Reports: Hold

```
1
2 *****
3 Report : timing
4     -path full
5     -delay min
6     -max_paths 20
7 Design : SYS_TOP
8 Version: K-2015.06
9 Date   : Sat Oct 26 06:57:38 2024
10 *****
11
12 Operating Conditions: scmetro_tsmc_cl013g_rvt_ff_1p32v_m40c  Library: scmetro_tsmc_cl013g_rvt_ff_1p32v_m40c
13 Wire Load Model Mode: top
14
15 Startpoint: DATA_SYNC_U0/enable_pulse_reg
16             (rising edge-triggered flip-flop clocked by REF_CLK)
17 Endpoint:  ALU_U0/OUT_VALID_reg
18             (rising edge-triggered flip-flop clocked by ALU_CLK)
19 Path Group: ALU_CLK
20 Path Type: min
21
22 Point                Incr      Path
23 -----
24 clock REF_CLK (rise edge)                0.00      0.00
25 clock network delay (ideal)              0.00      0.00
26 DATA_SYNC_U0/enable_pulse_reg/CK (DFFRQX2M) 0.00      0.00 r
27 DATA_SYNC_U0/enable_pulse_reg/Q (DFFRQX2M) 0.53      0.53 f
28 DATA_SYNC_U0/enable_pulse (DATA_SYNC_NUM_STAGES2_BUS_WIDTH8)
29                                     0.00      0.53 f
30 SYS_CTRL_U0/RX_D_VLD (SYS_CTRL)          0.00      0.53 f
31 SYS_CTRL_U0/U51/Y (NAND2X2M)             0.17      0.70 r
32 SYS_CTRL_U0/U38/Y (INVX2M)               0.09      0.79 f
33 SYS_CTRL_U0/ALU_EN (SYS_CTRL)            0.00      0.79 f
34 ALU_U0/EN (ALU_OPER_WIDTH8_OUT_WIDTH16) 0.00      0.79 f
35 ALU_U0/OUT_VALID_reg/D (DFFRQX2M)        0.00      0.79 f
36 data arrival time                        0.79
37
38 clock ALU_CLK (rise edge)                0.00      0.00
39 clock network delay (ideal)              0.00      0.00
40 clock uncertainty                        0.10      0.10

```

"violate" not found



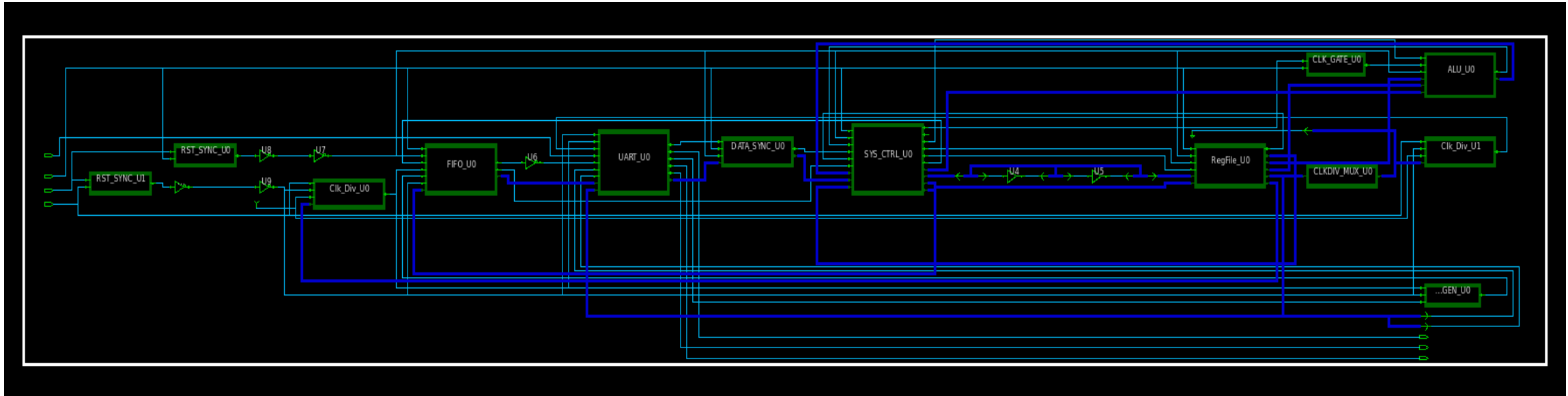
Area Report

```
1
2 *****
3 Report : area
4 Design : SYS_TOP
5 Version: K-2015.06
6 Date   : Sat Oct 26 06:57:37 2024
7 *****
8
9 Library(s) Used:
10
11   scmetro_tsmc_cl013g_rvt_ss_1p08v_125c (File: /home/IC/tsmc_fb_cl013g_sc/aci/sc-m/synopsys/scmetro_tsmc_cl013g_rvt_ss_1p08v_125c.db)
12
13 Number of ports:           754
14 Number of nets:           2596
15 Number of cells:          1822
16 Number of combinational cells: 1430
17 Number of sequential cells:  354
18 Number of macros/black boxes:  0
19 Number of buf/inv:         246
20 Number of references:      16
21
22 Combinational area:        13497.925938
23 Buf/Inv area:              944.890120
24 Noncombinational area:     8844.077091
25 Macro/Black Box area:      0.000000
26 Net Interconnect area:     undefined (No wire load specified)
27
28 Total cell area:           22342.003029
```

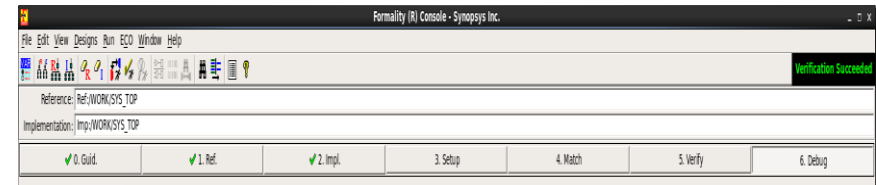
Power Report

```
1 Loading db file '/home/IC/tsmc_fb_cl013g_sc/aci/sc-m/synopsys/scmetro_tsmc_cl013g_rvt_ss_1p08v_125c.db'
2 Information: Propagating switching activity (low effort zero delay simulation). (PWR-6)
3 Warning: Design has unannotated primary inputs. (PWR-414)
4 Warning: Design has unannotated sequential cell outputs. (PWR-415)
5
6 *****
7 Report : power
8         -hier
9         -analysis_effort low
10 Design : SYS_TOP
11 Version: K-2015.06
12 Date   : Sat Oct 26 06:57:38 2024
13 *****
14
15
16 Library(s) Used:
17
18     scmetro_tsmc_cl013g_rvt_ss_1p08v_125c (File: /home/IC/tsmc_fb_cl013g_sc/aci/sc-m/synopsys/scmetro_tsmc_cl013g_rvt_ss_1p08v_125c.db)
19
20
21 Operating Conditions: scmetro_tsmc_cl013g_rvt_ss_1p08v_125c   Library: scmetro_tsmc_cl013g_rvt_ss_1p08v_125c
22 Wire Load Model Mode: top
23
24
25 Global Operating Voltage = 1.08
26 Power-specific unit information :
27     Voltage Units = 1V
28     Capacitance Units = 1.000000pf
29     Time Units = 1ns
30     Dynamic Power Units = 1mW   (derived from V,C,T units)
31     Leakage Power Units = 1pW
32
33
34 -----
35 Hierarchy      Switch  Int    Leak   Total
36                Power   Power  Power  Power  %
37 -----
38 SYS TOP        6.21e-03  0.194  1.35e+07  0.214  100.0
```

Netlist



Formal Verification



TCL script snippet

```
58 ##### set the top Reference Design #####
59
60 set_reference_design SYS_TOP
61 set_top SYS_TOP
62
63 ##### Read Implementation tech libs #####
64
65 read_db -container Imp [list $SSLIB $TTLIB $FFLIB]
66
67 ##### Read Implementation Design Files #####
68
69 read_verilog -container Imp -netlist "/home/IC/Projects/System/Backend/Synthesis/netlists/SYS_TOP.v"
70
71 ##### set the top Implementation Design #####
72
73 set_implementation_design SYS_TOP
74 set_top SYS_TOP
75
76
77 ## matching Compare points
78 match
79
80 ## verify
81 set successful [verify]
82 if {!$successful} {
83 diagnose
84 analyze_points -failing
85 }
```

Verification Result

Formality (R) Console - Synopsys Inc.

File Edit View Designs Run ECO Window Help

Verification Succeeded

Reference: Ref:/WORK/SYS_TOP
Implementation: Imp:/WORK/SYS_TOP

0. Guid. 1. Ref. 2. Impl. 3. Setup 4. Match 5. Verify 6. Debug

Failing Points Passing Points Aborted Points Unverified Points Probe Points Analyses Loops

	Type	Reference	Size	Implementation	Size	+/-
1	DFF	ALU_U0/ALU_OUT_reg[0]		ALU_U0/ALU_OUT_reg[0]		
2	DFF	ALU_U0/ALU_OUT_reg[10]		ALU_U0/ALU_OUT_reg[10]		
3	DFF	ALU_U0/ALU_OUT_reg[11]		ALU_U0/ALU_OUT_reg[11]		
4	DFF	ALU_U0/ALU_OUT_reg[12]		ALU_U0/ALU_OUT_reg[12]		
5	DFF	ALU_U0/ALU_OUT_reg[13]		ALU_U0/ALU_OUT_reg[13]		
6	DFF	ALU_U0/ALU_OUT_reg[14]		ALU_U0/ALU_OUT_reg[14]		
7	DFF	ALU_U0/ALU_OUT_reg[15]		ALU_U0/ALU_OUT_reg[15]		
8	DFF	ALU_U0/ALU_OUT_reg[1]		ALU_U0/ALU_OUT_reg[1]		
9	DFF	ALU_U0/ALU_OUT_reg[2]		ALU_U0/ALU_OUT_reg[2]		
10	DFF	ALU_U0/ALU_OUT_reg[3]		ALU_U0/ALU_OUT_reg[3]		
11	DFF	ALU_U0/ALU_OUT_reg[4]		ALU_U0/ALU_OUT_reg[4]		

of Passing Points: 344

Display names: Original Mapped Analyze Analyze Selected Points

DFT Insertion



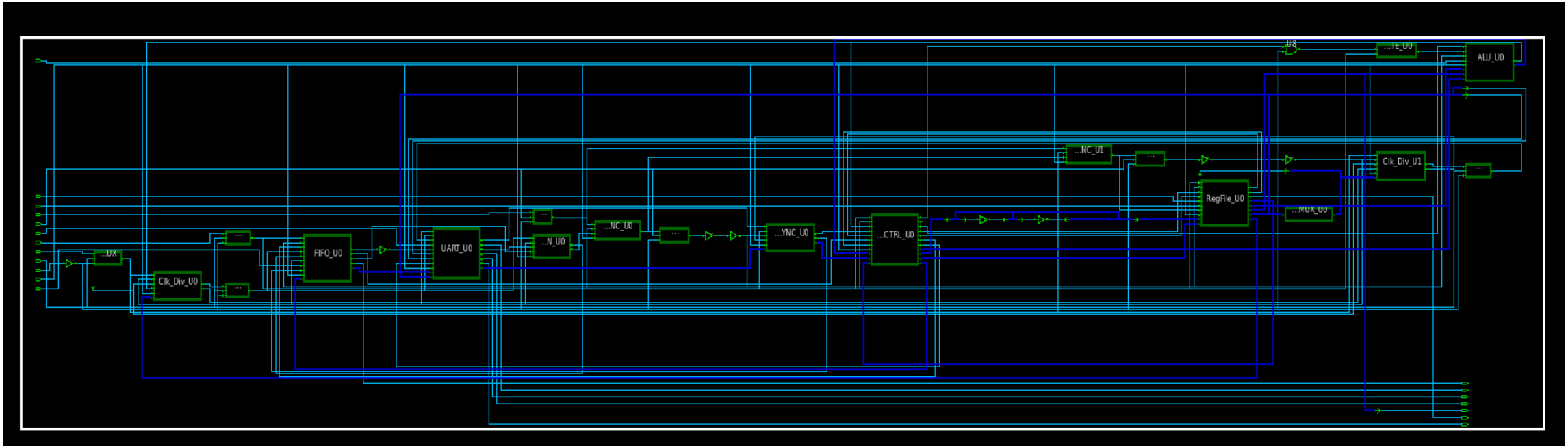
TCL script snippet

```
145
146 ##### Define DFT Signals #####
147
148 set_dft_signal -port [get_ports scan_clk] -type ScanClock -view existing_dft -timing {30 60}
149 set_dft_signal -port [get_ports scan_rst] -type Reset -view existing_dft -active_state 0
150 set_dft_signal -port [get_ports test_mode] -type Constant -view existing_dft -active_state 1
151 set_dft_signal -port [get_ports test_mode] -type TestMode -view spec -active_state 1
152 set_dft_signal -port [get_ports SE] -type ScanEnable -view spec -active_state 1 -usage scan
153 set_dft_signal -port [get_ports SI] -type ScanDataIn -view spec
154 set_dft_signal -port [get_ports SO] -type ScanDataOut -view spec
155
156
157 ##### Create Test Protocol #####
158
159 create_test_protocol
160
161 ##### Pre-DFT Design Rule Checking #####
162
163 dft_drc -verbose
164
165 ##### Preview DFT #####
166
167 preview_dft -show scan_summary
168
169 ##### Insert DFT #####
170
171 insert_dft
172
173 ##### Optimize Logic post DFT #####
174
175 compile -scan -incremental
176
177 ##### Design Rule Checking post DFT #####
178
179 dft_drc -verbose -coverage_estimate > reports/coverage.rpt
180
```

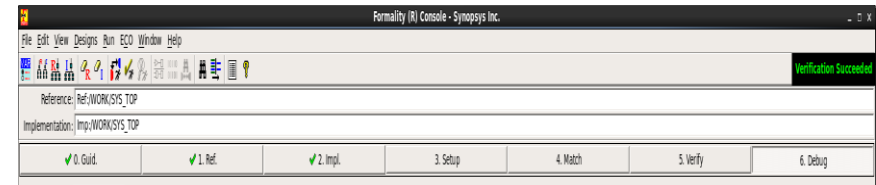

Coverage

```
427      Uncollapsed Stuck Fault Summary Report
428 -----
429 fault class          code    #faults
430 -----
431 Detected              DT      14998
432 Possibly detected     PT        0
433 Undetectable          UD       198
434 ATPG untestable      AU        51
435 Not detected          ND         9
436 -----
437 total faults              15256
438 test coverage           99.60%
439 -----
440 Information: The test coverage above may be inferior
441               than the real test coverage with customized
442               protocol and test simulation library.
443 1
```

Netlist



Formal Verification



TCL script snippet

```
58 read_sverilog -container Ref "SYS_TOP_dft.sv"
59
60 ##### set the top Reference Design #####
61
62 set_reference_design SYS_TOP
63 set_top SYS_TOP
64
65 ##### Read Implementation tech libs #####
66
67 read_db -container Imp [list $SSLIB $TTLIB $FFLIB]
68
69 ##### Read Implementation Design Files #####
70
71 read_verilog -container Imp -netlist "/home/IC/Projects/System/Backend/DFT/netlists/SYS_TOP.v"
72
73 ##### set the top Implementation Design #####
74
75 set_implementation_design SYS_TOP
76 set_top SYS_TOP
77
78 ##### Don't verify #####
79
80 # do not verify scan in & scan out ports as a compare point as it is existed only after synthesis and not existed in the RTL
81
82 #scan in
83 set_dont_verify_points -type port Ref:/WORK/*/SI
84 set_dont_verify_points -type port Imp:/WORK/*/SI
85
86 #scan out
87 set_dont_verify_points -type port Ref:/WORK/*/SO
88 set_dont_verify_points -type port Imp:/WORK/*/SO
```

Verification Result

Formality (R) Console - Synopsys Inc.

File Edit View Designs Run ECO Window Help

Verification Succeeded

Reference: Ref:/WORK/SYS_TOP
Implementation: Imp:/WORK/SYS_TOP

0. Guid. 1. Ref. 2. Impl. 3. Setup 4. Match 5. Verify 6. Debug

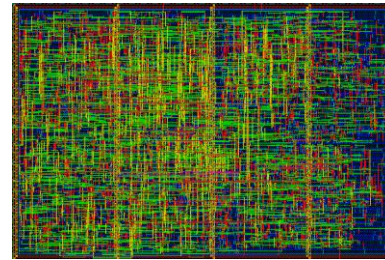
Failing Points	Passing Points	Aborted Points	Unverified Points	Probe Points	Analyses	Loops
Type	Reference	Size	Implementation	Size	+/-	
1	DFF	ALU_U0/ALU_OUT_reg[0]	ALU_U0/ALU_OUT_reg[0]			
2	DFF	ALU_U0/ALU_OUT_reg[10]	ALU_U0/ALU_OUT_reg[10]			
3	DFF	ALU_U0/ALU_OUT_reg[11]	ALU_U0/ALU_OUT_reg[11]			
4	DFF	ALU_U0/ALU_OUT_reg[12]	ALU_U0/ALU_OUT_reg[12]			
5	DFF	ALU_U0/ALU_OUT_reg[13]	ALU_U0/ALU_OUT_reg[13]			
6	DFF	ALU_U0/ALU_OUT_reg[14]	ALU_U0/ALU_OUT_reg[14]			
7	DFF	ALU_U0/ALU_OUT_reg[15]	ALU_U0/ALU_OUT_reg[15]			
8	DFF	ALU_U0/ALU_OUT_reg[1]	ALU_U0/ALU_OUT_reg[1]			
9	DFF	ALU_U0/ALU_OUT_reg[2]	ALU_U0/ALU_OUT_reg[2]			
10	DFF	ALU_U0/ALU_OUT_reg[3]	ALU_U0/ALU_OUT_reg[3]			
11	DFF	ALU_U0/ALU_OUT_reg[4]	ALU_U0/ALU_OUT_reg[4]			

of Passing Points: 344

Display names: ☐ Original ☒ Mapped

Analyze Analyze Selected Points

PnR Flow



FloorPlanning: Design Import

```
1 ##### Variables #####
2
3 set load_fp 0
4
5 ##### Define Top Module & Gate Level Netlist #####
6 set top_module SYS_TOP
7 set gate_level_netlist "/home/ahesham/Projects/System_pnr/DFT/netlists/SYS_TOP.v"
8
9 setUIVar rda_Input ui_topcell $top_module
10 setUIVar rda_Input ui_netlist $gate_level_netlist
11
12 ##### Define Tech Libraries Timing Views #####
13 set FFLIB "/home/ahesham/Projects/System_pnr/std_cells/libs/scmetro_tsmc_cl013g_rvt_ff_1p32v_m40c.lib"
14 set SSLIB "/home/ahesham/Projects/System_pnr/std_cells/libs/scmetro_tsmc_cl013g_rvt_ss_1p08v_125c.lib"
15 set TTLIB "/home/ahesham/Projects/System_pnr/std_cells/libs/scmetro_tsmc_cl013g_rvt_tt_1p2v_25c.lib"
16
17 setUIVar rda_Input ui_timelib,min $FFLIB
18 setUIVar rda_Input ui_timelib,max $SSLIB
19 setUIVar rda_Input ui_timelib $TTLIB
20
21 ##### Define Tech Libraries LEF Views #####
22 set TECH_LEF "/home/ahesham/Projects/System_pnr/std_cells/lef/tsmc13fsg_7lm_tech.lef"
23 set MACRO_LEF "/home/ahesham/Projects/System_pnr/std_cells/lef/tsmc13_m_macros.lef"
24 set DESIGN_LEF "/home/ahesham/Projects/System_pnr/pnr/import/SYS_TOP.lef"
25
26 if {$load_fp == 0} {
27
28 setUIVar rda_Input ui_leffile [list $TECH_LEF $MACRO_LEF $DESIGN_LEF]
29
30 } else {
31
32 setUIVar rda_Input ui_leffile [list $TECH_LEF $MACRO_LEF]
33
34 }
35 ##### Define Capacitance Table #####
36 set CAP_TABLE_FILE "/home/ahesham/Projects/System_pnr/std_cells/captables/tsmc13fsg.capTbl"
37
38 setUIVar rda_Input ui_captbl_file $CAP_TABLE_FILE
39
40 ##### Define Constraints File #####
```



FloorPlanning: Multi Mode Multi Corner

```
1
2 create_library_set -name min_library -timing "../std_cells/libs/scmetro_tsmc_cl013g_rvt_ff_1p32v_m40c.lib"
3 create_library_set -name max_library -timing "../std_cells/libs/scmetro_tsmc_cl013g_rvt_ss_1p08v_125c.lib"
4 create_library_set -name typ_library -timing "../std_cells/libs/scmetro_tsmc_cl013g_rvt_tt_1p2v_25c.lib"
5
6 create_constraint_mode -name func_mode -sdcs_files {../DFT/sdc/SYS_TOP_func.sdc}
7 create_constraint_mode -name scan_mode -sdcs_files {../DFT/sdc/SYS_TOP_scan.sdc}
8 create_constraint_mode -name capture_mode -sdcs_files {../DFT/sdc/SYS_TOP_capture.sdc}
9
10 create_rc_corner -name RCcorner -cap_table "../std_cells/captables/tsmc13fsg.capTbl"
11
12 create_delay_corner -name min_corner -library_set min_library -rc_corner RCcorner
13 create_delay_corner -name max_corner -library_set max_library -rc_corner RCcorner
14
15 #func
16 create_analysis_view -name setup1_analysis_view -delay_corner max_corner -constraint_mode func_mode
17 create_analysis_view -name hold1_analysis_view -delay_corner min_corner -constraint_mode func_mode
18 #capture
19 create_analysis_view -name setup2_analysis_view -delay_corner max_corner -constraint_mode capture_mode
20 create_analysis_view -name hold2_analysis_view -delay_corner min_corner -constraint_mode capture_mode
21
22 #scan
23 create_analysis_view -name setup3_analysis_view -delay_corner max_corner -constraint_mode scan_mode
24 create_analysis_view -name hold3_analysis_view -delay_corner min_corner -constraint_mode scan_mode
25
26 set_analysis_view -setup {setup1_analysis_view setup2_analysis_view setup3_analysis_view} \
27 -hold {hold1_analysis_view hold2_analysis_view hold3_analysis_view}
```



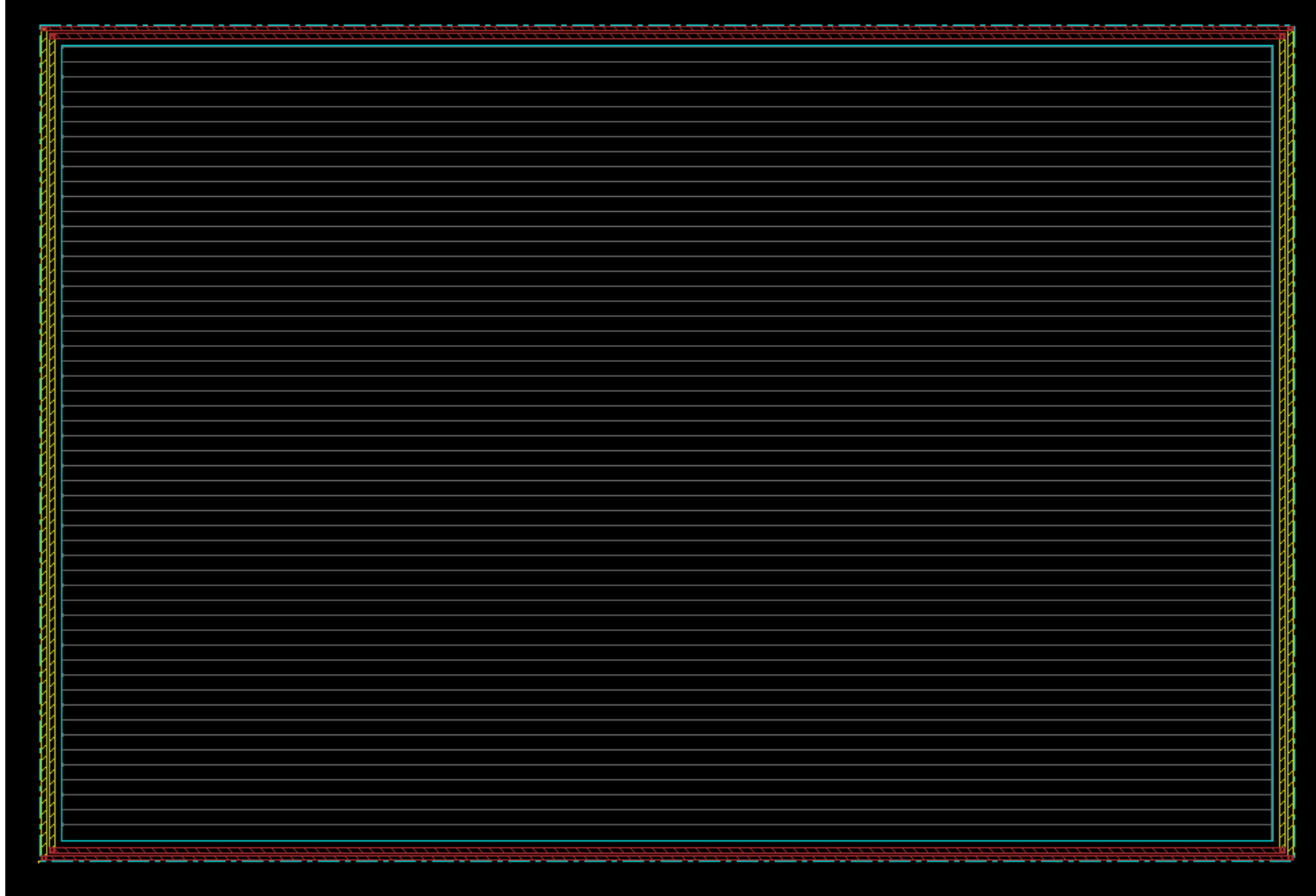
FloorPlanning: TCL script

```
2
3 ##### Variables #####
4
5 set load_fp 0
6 set top_module SYS_TOP
7
8 ##### Define Aspect Ratio (Length/Width) of Digital Macro #####
9
10 if {$load_fp == 0} {
11
12 floorPlan -d 240.47 160.47 4.0 4.0 4.0 4.0
13
14
15 } else {
16
17 loadFPlan ./$top_module.fp
18
19 }
20
21
```

FloorPlanning: Schematic



Power Planning: Power Rings



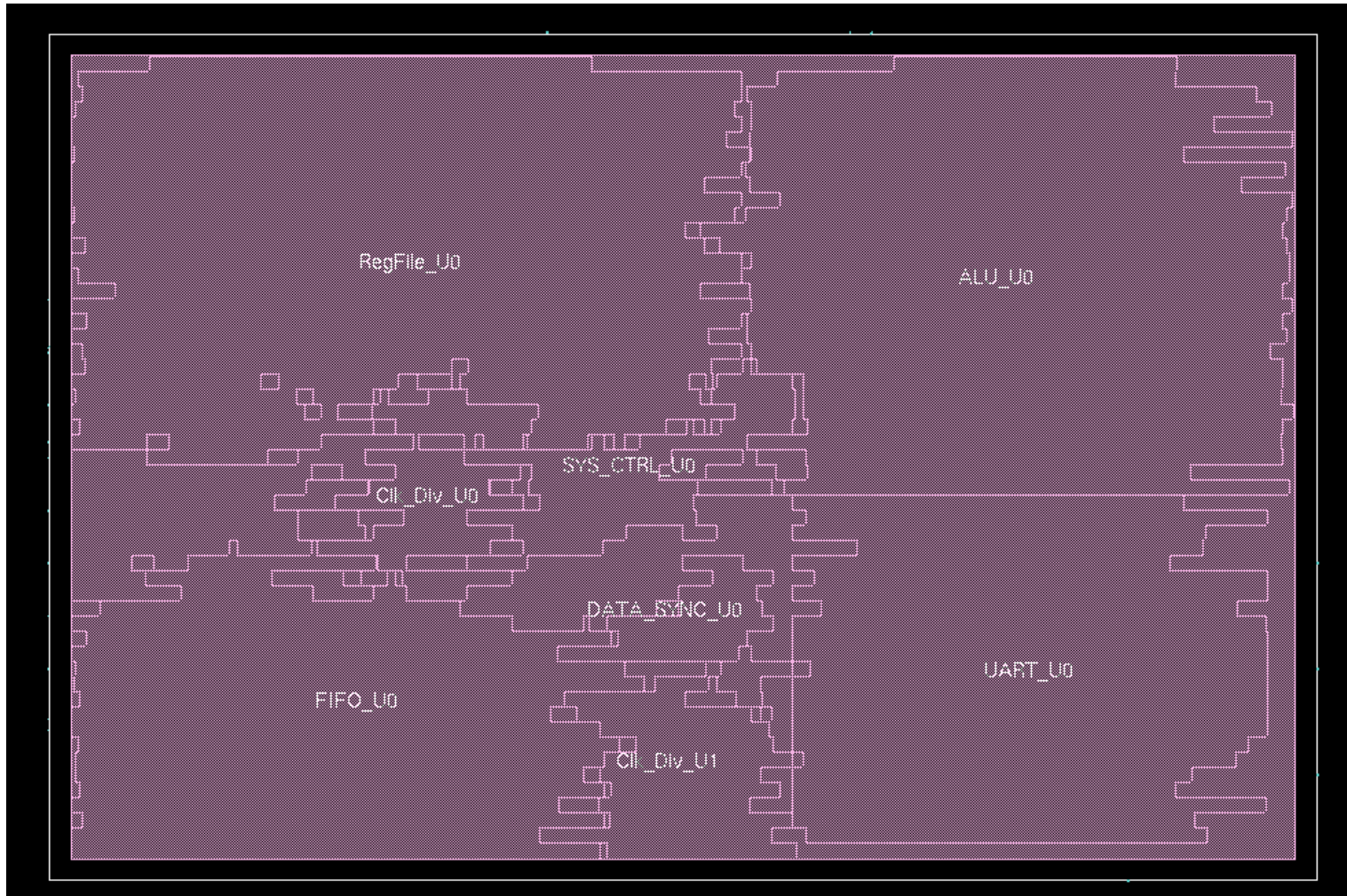
Power Planning: Power Stripes



Placement: TCL Script

```
2 ##### Placement Standard Cells - TIE High/Low -- connect VDD/VSS #####
3 placeDesign -inPlaceOpt -prePlaceOpt
4 addTieHiLo -cell TIELOM -prefix LTIE
5 addTieHiLo -cell TIEHIM -prefix HTIE
6 globalNetConnect VDD -type pgpin -pin VDD -inst *
7 globalNetConnect VSS -type pgpin -pin VSS -inst *
```

Placement: Schematic



Setup Analysis

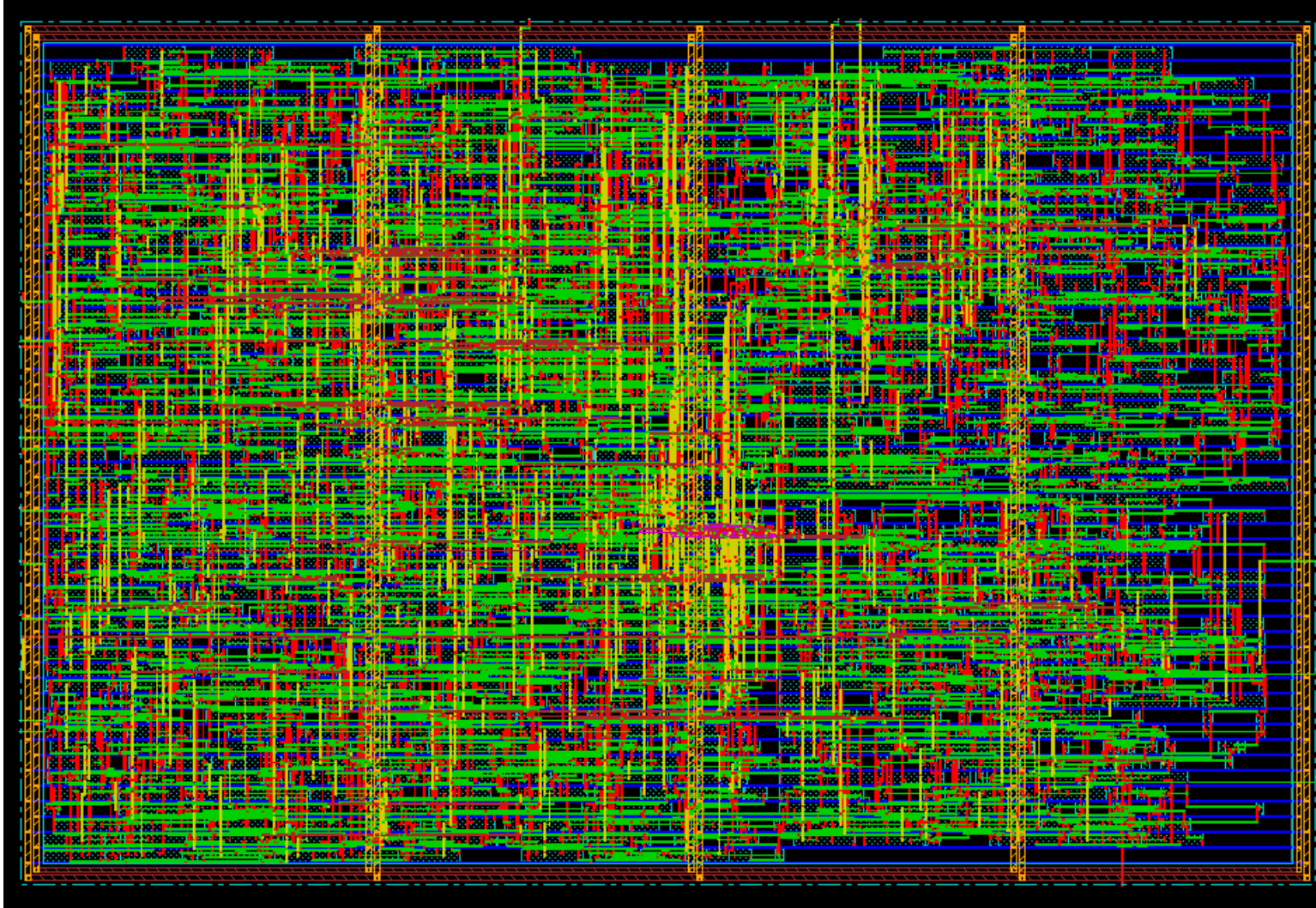
```
-----
timeDesign Summary
-----

+-----+-----+-----+-----+-----+-----+
| Setup mode | all | reg2reg | in2reg | reg2out | in2out | clkgate |
+-----+-----+-----+-----+-----+-----+
| WNS (ns): | 2.449 | 2.449 | 16.022 | 38.783 | N/A | 18.271 |
| TNS (ns): | 0.000 | 0.000 | 0.000 | 0.000 | N/A | 0.000 |
| Violating Paths: | 0 | 0 | 0 | 0 | N/A | 0 |
| All Paths: | 1344 | 1010 | 680 | 4 | N/A | 1 |
+-----+-----+-----+-----+-----+-----+

+-----+-----+-----+
| DRV's | Real | Total |
|-----+-----+-----|
| | Nr nets(terms) | Worst Vio | Nr nets(terms) |
+-----+-----+-----+
| max_cap | 0 (0) | 0.000 | 0 (0) |
| max_tran | 0 (0) | 0.000 | 0 (0) |
| max_fanout | 0 (0) | 0 | 0 (0) |
+-----+-----+-----+

Density: 73.922%
Routing Overflow: 0.00% H and 0.37% V
-----
Reported timing to dir timingReports
Total CPU time: 1.27 sec
Total Real time: 1.0 sec
Total Memory Usage: 318.421875 Mbytes
encounter 1> █
```


Clock Tree Synthesis



Setup Analysis

timeDesign Summary

Setup mode	all	reg2reg	in2reg	reg2out	in2out	clkgate
WNS (ns):	2.403	2.403	16.255	37.421	N/A	17.829
TNS (ns):	0.000	0.000	0.000	0.000	N/A	0.000
Violating Paths:	0	0	0	0	N/A	0
All Paths:	1344	1010	680	4	N/A	1

DRVs	Real		Total
	Nr nets(terms)	Worst Vio	Nr nets(terms)
max_cap	0 (0)	0.000	0 (0)
max_tran	0 (0)	0.000	0 (0)
max_fanout	0 (0)	0	0 (0)

Density: 73.869%

Routing Overflow: 0.00% H and 0.37% V

Reported timing to dir timingReports

Total CPU time: 1.26 sec

Total Real time: 1.0 sec

Total Memory Usage: 319.863281 Mbytes

encounter 2> █

Hold Analysis

```
-----
timeDesign Summary
-----

+-----+-----+-----+-----+-----+-----+
| Hold mode | all | reg2reg | in2reg | reg2out | in2out | clkgate |
+-----+-----+-----+-----+-----+-----+
|      WNS (ns): | 0.041 | 0.065 | 0.041 | 10.886 | N/A | 0.498 |
|      TNS (ns): | 0.000 | 0.000 | 0.000 | 0.000 | N/A | 0.000 |
| Violating Paths: | 0 | 0 | 0 | 0 | N/A | 0 |
| All Paths: | 1344 | 1010 | 680 | 4 | N/A | 1 |
+-----+-----+-----+-----+-----+-----+

Density: 73.922%
Routing Overflow: 0.00% H and 0.37% V
-----
Reported timing to dir timingReports
Total CPU time: 1.47 sec
Total Real time: 1.0 sec
Total Memory Usage: 322.398438 Mbytes
encounter 2> █
```

Routing: TCL Script

```
1 |
2 | ##### Variables #####
3 | set route_0_optimize_1 1
4 |
5 | ##### Global/Detailed Routing #####
6 | set max_m_layer 6
7 |
8 | if {$route_0_optimize_1 == 0} {
9 |   setNanoRouteMode -quiet -routeTopRoutingLayer $max_m_layer
10 |  setNanoRouteMode -quiet -routeBottomRoutingLayer default
11 |  setNanoRouteMode -quiet -drouteEndIteration default
12 |  setNanoRouteMode -quiet -routeWithTimingDriven false
13 |  setNanoRouteMode -quiet -routeWithSiDriven false
14 |  setNanoRouteMode -quiet -routeWithSiDriven true
15 |  setNanoRouteMode -quiet -routeSiEffort max
16 |
17 |  routeDesign -globalDetail -viaOpt -wireOpt
18 |
19 | } else {
20 |
21 |  refinePlace -preserveRouting
22 |  setNanoRouteMode -routeWithEco true
23 |  setNanoRouteMode -droutePostRouteSwapVia true
24 |  globalDetailRoute
25 |
26 | }
```

Verification: Connectivity

```
encounter 3>
***** Start: VERIFY CONNECTIVITY *****
Start Time: Sat Oct 26 15:37:40 2024

Design Name: SYS_TOP
Database Units: 2000
Design Boundary: (0.0000, 0.0000) (240.4700, 160.4700)
Error Limit = 1000; Warning Limit = 50
Check all nets
Net RST_N: Found a geometry with bounding box (-0.10,70.10) (0.10,70.30) outside design boundary.
Violations for such geometries will be reported.

Begin Summary
  Found no problems or warnings.
End Summary

End Time: Sat Oct 26 15:37:40 2024
***** End: VERIFY CONNECTIVITY *****
  Verification Complete : 0 Viols.  0 Wrngs.
  (CPU Time: 0:00:00.2  MEM: 0.000M)

encounter 3> █
```

Verification: Geomtry

```
encounter 3> *** Starting Verify Geometry (MEM: 379.9) ***

VERIFY GEOMETRY ..... Starting Verification
VERIFY GEOMETRY ..... Initializing
VERIFY GEOMETRY ..... Deleting Existing Violations
VERIFY GEOMETRY ..... Creating Sub-Areas
                      ..... bin size: 8000
VERIFY GEOMETRY ..... SubArea : 1 of 1
VERIFY GEOMETRY ..... Cells      : 0 Viols.
VERIFY GEOMETRY ..... SameNet    : 0 Viols.
VERIFY GEOMETRY ..... Wiring     : 0 Viols.
VERIFY GEOMETRY ..... Antenna    : 0 Viols.
VERIFY GEOMETRY ..... Sub-Area : 1 complete 0 Viols. 0 Wrngs.
VG: elapsed time: 1.00
Begin Summary ...
Cells      : 0
SameNet    : 0
Wiring     : 0
Antenna    : 0
Short      : 0
Overlap    : 0
End Summary

Verification Complete : 0 Viols. 0 Wrngs.

*****End: VERIFY GEOMETRY*****
*** verify geometry (CPU: 0:00:01.0 MEM: 28.4M)

encounter 3> █
```

Setup Analysis

```
-----
timeDesign Summary
-----

+-----+-----+-----+-----+-----+-----+-----+
| Setup mode | all | reg2reg | in2reg | reg2out | in2out | clkgate |
+-----+-----+-----+-----+-----+-----+-----+
| WNS (ns): | 2.750 | 2.750 | 16.346 | 37.454 | N/A | 17.862 |
| TNS (ns): | 0.000 | 0.000 | 0.000 | 0.000 | N/A | 0.000 |
| Violating Paths: | 0 | 0 | 0 | 0 | N/A | 0 |
| All Paths: | 1344 | 1010 | 680 | 4 | N/A | 1 |
+-----+-----+-----+-----+-----+-----+-----+

+-----+-----+-----+-----+
| DRV's | Real | Total |
|-----+-----+-----|
| | Nr nets(terms) | Worst Vio | Nr nets(terms) |
+-----+-----+-----+
| max_cap | 0 (0) | 0.000 | 0 (0) |
| max_tran | 0 (0) | 0.000 | 0 (0) |
| max_fanout | 0 (0) | 0 | 0 (0) |
+-----+-----+-----+

Density: 73.922%
-----
Reported timing to dir timingReports
Total CPU time: 1.35 sec
Total Real time: 2.0 sec
Total Memory Usage: 388.421875 Mbytes
encounter 3> █
```

Hold Analysis

```
-----
timeDesign Summary
-----

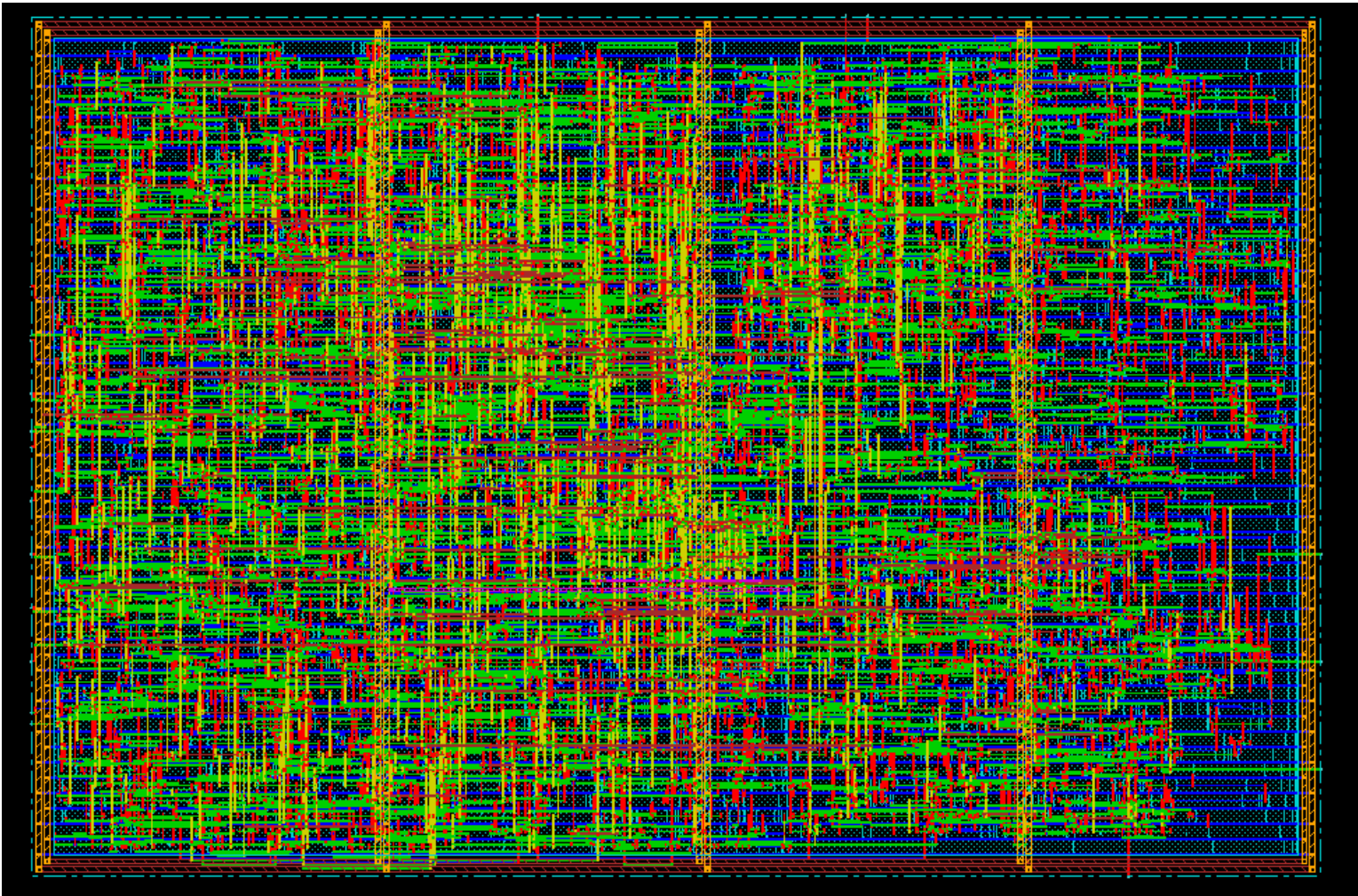
+-----+-----+-----+-----+-----+-----+-----+
| Hold mode | all | reg2reg | in2reg | reg2out | in2out | clkgate |
+-----+-----+-----+-----+-----+-----+-----+
|      WNS (ns): | 0.047 | 0.062 | 0.047 | 10.874 | N/A | 0.492 |
|      TNS (ns): | 0.000 | 0.000 | 0.000 | 0.000 | N/A | 0.000 |
| Violating Paths: | 0 | 0 | 0 | 0 | N/A | 0 |
|    All Paths: | 1344 | 1010 | 680 | 4 | N/A | 1 |
+-----+-----+-----+-----+-----+-----+-----+

Density: 73.922%
-----
Reported timing to dir timingReports
Total CPU time: 1.29 sec
Total Real time: 2.0 sec
Total Memory Usage: 388.421875 Mbytes
encounter 3> █
```

Chip Finishing: TCL Script

```
2 set top_cell ALU_TOP
3
4 # Add Filler Cells
5 set filler_cell_list {FILL1M FILL2M FILL4M FILL8M FILL16M FILL32M FILL64M}
6 addFiller -cell $filler_cell_list -prefix FILLER -markFixed
7
8
```


Chip Finishing: Schematic



Generated Files



SYS_TOP.gds



SYS_TOP.sdf



SYS_TOP.spf



SYS_TOP.v



SYS_TOP_pg.v



Thank you

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